

Nelson Roque

The Pennsylvania State University
Department of Human Development and Family Studies
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Appointments

2023–Present — Assistant Professor

Department of Human Development and Family Studies, Penn State University

Affiliations: Social Science Research Institute (SSRI); Geroscience Consortium, Center for Healthy Aging

2020–2023 — Assistant Professor

Department of Psychology, Human Factors and Cognitive Psychology Program, University of Central Florida

Career Highlights

Metric	Count / Amount	Metric	Count / Amount
Peer-reviewed publications	47	Books / chapters	4
Current external funding	5	Total external funding	\$23,915,607
Invited talks	7	NIH grants	8
Program leadership roles	3	NIH reviewer service	2
Major media coverage	26		

Education

1. 2020 — National Institute on Aging (NIA) T32 Postdoctoral Fellow, Pathways T32 Training Program. Center for Healthy Aging, Pennsylvania State University
2. 2018 — PhD, Cognitive Psychology, Florida State University
3. 2015 — M.S., Cognitive Psychology, Florida State University
4. 2012 — B.S., Psychology, Florida State University

Research Areas

Cognitive Aging, Digital Health, Environmental Health, Human Factors, AI and Aging, Mobile Cognitive Assessment

Peer-Reviewed Articles

Supervised students and postdocs are underlined.

1. **Roque, N. A.**, Hosgood, H. D., & Hall, C. B. (2026). The importance of accurate indoor air quality assessment in the aging and dementia population. *Journal of Alzheimer's Disease*, 109(2), 630-632, 13872877251396928. <https://doi.org/10.1177/13872877251396928>
2. Shaleha, R., **Roque, N.**, Andrews, H., Calfee, K., & Lee, S. (2026). *Screen use measurement tools: A mapping of instruments, gaps, and future directions*. *Cyberpsychology, Behavior, and Social Networking**, 29(3), 140–150.
3. Shaleha, R. R. A., & **Roque, N. A.** (2026). Screen time in context: Toward a theoretical model of digital engagement across the lifespan. *Developmental Psychology*. Advance online publication. <https://doi.org/10.1037/dev0002202>
4. **Roque, N. A.**, Andrews, H., & Santos-Lozada, A. R. (2025). Identifying air quality monitoring deserts in the United States. *Proceedings of the National Academy of Sciences*, 122(17), e2425310122. <https://doi.org/10.1073/pnas.2425310122>
5. Doheny, M. M., **Roque, N. A.**, & Lighthall, N. R. (2025). Effects of maximum and minimum offers on reciprocity and trust perceptions during economic decision-making. *Frontiers in Cognition*, 4. Fcogn.2025.1576987. <https://doi.org/10.3389/fcogn.2025.1576987>
6. Katz, M. J., Johns, T. S., Martin, M. M., Burgess, A., Claris, V., Volda, G. S., **Roque, N.**, Shaw, P. A., Hyun, N., Hakun, J. G., Pavlovic, J. M., Wylie-Rosett, J., Sliwinski, M. J., & Mossavar-Rahmani, Y. (2025). Recruitment of mid-life adults to a randomized clinical trial: The multicultural healthy diet study to reduce cognitive decline and Alzheimer's disease risk. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*, 11(4), e70174. <https://doi.org/10.1002/trc2.70174>
7. **Roque, N.**, & Felt, J. (2025). Acceptability of Active and Passive Data Collection Methods for Mobile Health Research: Cross-Sectional Survey of an Online Adult Sample in the United States. *JMIR Formative Research*, 9(1), e64082. <https://doi.org/10.2196/64082>
8. Shaleha, R., & **Roque, N.** (2025). From screens to cognition: A scoping review of the impact of screen time on cognitive function in midlife and older adults. *Digital health*, 11, 20552076251343989. <https://doi.org/10.1177/20552076251343989>
9. Oravec, Z., Vandekerckhove, J., Hakun, J.G., Kim, S.H., Katz, M.J., Wang, C., Lipton, R.B., Derby, C.A., **Roque, N.A.** and Sliwinski, M.J., (2025). Computational phenotyping of cognitive decline with retest learning. *The Journals of Gerontology, Series B: Psychological Sciences and Social Sciences*, 80(7), gbaf030. <https://doi.org/10.1093/geronb/gbaf030>
10. García de la Garza, Á., Nester, C., Wang, C., Mogle, J., **Roque, N.**, Katz, M., CA Derby, RB Lipton, & Rabin, L. (2025). Enhanced associations between subjective cognitive concerns and blood-based AD biomarkers using a novel EMA approach. *Alzheimer's Research & Therapy*, 17(1), 82, s13195-025-01720-y. <https://doi.org/10.1186/s13195-025-01720-y>
11. Nester, C. O., De Vito, A. N., Prieto, S., Kunicki, Z. J., Strenger, J., Harrington, K.

- D., **Roque, N.A.**, Sliwinski, M.J., Rabin, L.A., & Thompson, L. I. (2025). Association of Subjective Cognitive Concerns With Performance on Mobile App-Based Cognitive Assessment in Cognitively Normal Older Adults: Observational Study. *JMIR Aging*, 8, e64033. <https://doi.org/10.2196/64033>
12. Shaleha, R.R.A., & **Roque, N.** (2024). *Cognitive Performance in the Digital Era: Generational Differences, Stress, and Distraction's Impact on Cognitive Performance*. International Journal of Cyber Behavior, Psychology and Learning (IJCBL)*, 14(1), 1-13, IJCBL.341788. <https://doi.org/10.4018/IJCBL.341788>
 13. Adamo, S.H., **Roque, N.**, Barufaldi, B., Schmidt, J., Mello-Thoms, C., & Lago, M. (2023). *Assessing satisfaction of search in virtual mammograms for experienced and novice searchers*. Journal of Medical Imaging*, 10(S1), S11917-S11917. <https://doi.org/10.1117/1.JMI.10.S1.S11917>
 14. Harrell, E. R., **Roque, N.A.**, & Boot, W. R. (2023). Comparing the effectiveness of two theory-based strategies to promote cognitive training adherence. *Journal of Experimental Psychology: Applied*, 29(4), 782, xap0000485. <https://doi.org/10.1037/xap0000485>
 15. Hyun, J., Katz, M. J., Derby, C. A., **Roque, N.**, Muñoz, E., Sliwinski, M. J., Lovasi G. S. & Lipton, R. B. (2023). Availability of healthy foods, fruit and vegetable consumption, and cognition among urban older adults. *BMC geriatrics*, 23(1), 302, s12877-023-04003-z. <https://doi.org/10.1186/s12877-023-04003-z>
 16. Thompson, L.I., Kunicki, Z.J., Emrani, S., Strenger, J., De Vito, A.N., Britton, K.J., Dion, C., Harrington, K.D., **Roque, N.**, Salloway, S. and Sliwinski, M.J., (2023). Remote and in-clinic digital cognitive screening tools outperform the MoCA to distinguish cerebral amyloid status among cognitively healthy older adults. *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring*, 15(4), e12500. <https://doi.org/10.1002/dad2.12500>
 17. Hakun, J. G., **Roque, N.A.**, Gerver, C.R., & Cerino, E.S. (2023). Ultra-brief Assessment of Working Memory Capacity: Ambulatory Assessment Study Using Smartphones. *JMIR Formative Research*, 7, e40188. <https://doi.org/10.2196/40188>
 18. Wan, X., Lighthall, N. R., & **Roque, N.** (2024). *Predictors of Technology Use: Evidence from Elite Older Adults*. Activities, Adaptation & Aging*, 48(4), 648-669, 01924788.2023.2282276. <https://doi.org/10.1080/01924788.2023.2282276>
 19. Harrington, K. D., **Roque, N.A.**, Strenger, J., Correia, S., Salloway, S. P., Sliwinski, M. J., & Thompson, L. I. (2022). Loneliness and subjective cognitive impairment in preclinical Alzheimer's disease and healthy aging. *Alzheimer's & Dementia*, 18, e069279. <https://doi.org/10.1002/alz.069279>
 20. Nicosia J., Wang B., Aschenbrenner A.J., Sliwinski, M.J., Yabiku, S., **Roque, N.A.**, Germine, L.T., Bateman, R.J., Morris, J.C., Hassenstab, J. (2022). To BYOD or not: Are device latencies important for bring-your-own-device (BYOD) smartphone cognitive testing? *Behavior research methods*, 55(6), 2800-2812, s13428-022-01925-1. <https://doi.org/10.3758/s13428-022-01925-1>

21. Johnston, E. A., **Roque, N.A.**, Cole, B. H., Flanagan, M. P., Kris-Etherton, P. M., & Petersen, K. S. (2022). A Pilot Study of the Effect of Evening Almond Butter Consumption on Overnight and Fasting Interstitial Glucose. *Diabetology*, 3(4), 502-513, 3040038. <https://doi.org/10.3390/diabetology3040038> [**Premier Diabetes Journal - Chosen for Cover Story**]
22. Hoogendoorn, C. J., Qin, J., Wang, C., **Roque, N.**, Laurenceau, J. P., Katz, M. J., Derby, C. A., Lipton, R. B., & Gonzalez, J. S. (2022). Depressive symptoms mediate the relationship between diabetes and cognitive performance in a community-based sample of older adults. *Journal of Diabetes and its Complications*, 36(7), 108183. <https://doi.org/10.1016/j.jdiacomp.2022.108183>
23. Thompson, L. I., Harrington, K. D., **Roque, N.**, Strenger, J., Correia, S., Jones, R. N., Salloway, S., & Sliwinski, M. J. (2022). A highly feasible, reliable, and fully remote protocol for mobile app-based cognitive assessment in cognitively healthy older adults. *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring*, 14(1), e12283. <https://doi.org/10.1002/dad2.12283>
24. Pasquini, G., Ferguson, G., Bouklas, I., Vu, H., Zamani, M., Zhaoyang, R., Harrington, K.D., **Roque, N.A.**, Mogle, J., Schwartz, H.A., & Scott, S. B. (2022). The where and when of COVID-19: Using ecological and Twitter-based assessments to examine impacts in a temporal and community context. *PloS one*, 17(2), e0264280. <https://doi.org/10.1371/journal.pone.0264280> [**Premier Open Access Journal; IF = 3.75**]
25. He, Z., Tian, S., Singh, A., Chakraborty, S., Zhang, S., Lustria, M. L. A., Charness, N., **Roque, N.A.**, Harrell, E. & Boot, W. R. (2022). A Machine-Learning-Based Approach for Predicting Older Adults' Adherence to Technology-Based Cognitive Training. *Information Processing & Management*, 59(5), 103034. <https://doi.org/10.1016/j.ipm.2022.103034> [**Premier International Journal of Information Sciences; IF = 6.222**]
26. Harrell, E. R., **Roque, N.A.**, Boot, W. R., & Charness, N. (2021). Investigating message framing to improve adherence to technology-based cognitive interventions. *Psychology and aging*, 36(8), 974, pag0000629. <https://doi.org/10.1037/pag0000629> [**Premier Adult Development and Aging Journal; IF = 3.34**]
27. Yang, C. H., Hakun, J. G., **Roque, N.**, Sliwinski, M. J., & Conroy, D. E. (2021). Mindful walking and cognition in older adults: A proof of concept study using in-lab and ambulatory cognitive measures. *Preventive Medicine Reports*, 23, 101490. <https://doi.org/10.1016/j.pmedr.2021.101490>
28. Gray, N., Yoon, J. S., Charness, N., Boot, W. R., **Roque, N.A.**, Andringa, R., Harrell, E. R., Lewis, K. G., & Vitale, T. (2021). Relative effectiveness of general versus specific cognitive training for aging adults. *Psychology and aging*, 37(2), 210-221, pag0000663. <https://psycnet.apa.org/doi/10.1037/pag0000663> [**Premier Adult Development and Aging Journal; IF = 3.34**]
29. Cerino, E.S., Katz, M.J., Wang, C., Qin, J., Gao, Q., Hyun, J., Hakun, J.G., **Roque, N.A.**,

- Derby, C.A., Lipton, R.B., & Sliwinski, M.J. (2021). Variability in cognitive performance on mobile devices is sensitive to mild cognitive impairment: Results from the Einstein Aging Study. *Frontiers in Digital Health*, 3, 758031. <https://doi.org/10.3389/fdgth.2021.758031>
30. **Roque, N.A.**, & Boot, W. R. (2020). A New Tool for Assessing Older Adults' Wireless Network Proficiency: The Wireless Network Proficiency Questionnaire. *Journal of Applied Gerontology*, 40(5), 541-546, 0733464820935000 <https://doi.org/10.1177/0733464820935000>
 31. Emihovich, B., **Roque, N.A.**, & Mason, J. (2020). Can Video Gameplay Improve Undergraduates' Problem-Solving Skills?. *International Journal of Game-Based Learning (IJGBL)*, 10(2), 1-18, 2020040102. <https://doi.org/10.4018/ijgbl.2020040102>
 32. Lebeau, J.-C., Mason, J. R., **Roque, N.A.**, Tenenbaum, G. (2020). The effects of acute exercise on driving and executive functions in healthy older adults. *International Journal of Sport and Exercise Psychology*, 20(1), 283-301. <https://doi.org/10.1080/1612197X.2020.1849353>
 33. Mason, J. R., Tenenbaum, G., Jaime, S., **Roque, N.A.**, Maharaj, A., Figueroa, A. (2020). Arterial Stiffness and Cardiorespiratory Fitness Are Associated With Cognitive Function in Older Adults. *Behavioral Medicine*: 1-12, 48(1), 54-65, 08964289.2020.1825921. <https://doi.org/10.1080/08964289.2020.1825921>
 34. Boot, W. R., Andringa, R., Harrell, E. R., Dieciuc, M. A., & **Roque, N. A.** (2020). Older adults and video gaming for leisure: Lessons from the Center for Research and Education on Aging and Technology Enhancement (CREATE). *Gerontechnology*, 19(2), gt.2020.19.2.006.00. <https://doi.org/10.4017/gt.2020.19.2.006.00>
 35. **Roque, N.A.**, Ram, N. (2019). tsfeaturex: An R Package for Automating Time Series Feature Extraction. *Journal of Open Source Software*, 4(37), 1279, <https://doi.org/10.21105/joss.01279>
 36. Yoon, J. S., **Roque, N.A.**, Andringa, R., Harrell, E. R., Lewis, K. G., Vitale, T., Charness, N., & Boot, W. R. (2019). Intervention comparative effectiveness for adult cognitive training (ICE-ACT) trial: rationale, design, and baseline characteristics. *Contemporary Clinical Trials*, 78, 76-87, 2019.01.014. <https://doi.org/10.1016/j.cct.2019.01.014>. [**Premier International Clinical Trials Journal; IF=2.226**]
 37. Dieciuc, M., **Roque, N.A.**, Boot, W.R. (2019). The spatial dynamics of mouse-tracking reveal that attention capture is stimulus-driven rather than contingent upon top-down goals. *Journal of Experimental Psychology: Human Perception and Performance*, 45(10), 1285, xhp0000671. <https://doi.org/10.1037/xhp0000671>
 38. Souders, D. J., Charness, N., **Roque, N.A.**, & Pham, H. (2019). Aging: Older Adults' Driving Behavior Using Longitudinal and Lateral Warning Systems. *Human Factors*, 62(2), 229-248, 0018720819864510. <https://doi.org/10.1177/0018720819864510> [**Premier Human Factors Journal; IF=3.165**]
 39. Tinker, A., Fozard, J., Kearns, W., Mihailidis, A., Atoyebi, O., Bailey, A., Bet, P., Lapierre, N., Neubauer, N., **Roque, N.A.**, Vermeer, Y., Taipale, V. (2018). Twelve

- years of ISG masterclasses: Past, Present, and Future. *Gerontechnology: international journal on the fundamental aspects of technology to serve the ageing society*, 17(4), 232, gt.2018.17.4.004.00. <https://doi.org/10.4017/gt.2018.17.4.004.00> – Note, author list for junior scientists organized alphabetically.
40. Wright, T. J., **Roque, N.A.**, Boot, W. R., & Stothart, C. (2018). Attention capture, processing speed, and inattention blindness. *Acta Psychologica*, 190, 72-77, 2018.07.005. <https://doi.org/10.1016/j.actpsy.2018.07.005>
 41. Andringa, R., Boot, W.R., **Roque, N.A.**, & Ponnaluri, S. (2018). Hand proximity effects are fragile: a useful null result. *Cognitive Research: Principles and Implications*, 3(1), 7, s41235-018-0094-7. <https://doi.org/10.1186/s41235-018-0094-7>
 42. Multilab Direct Replication Team, including Boot, W.R., Andringa, R., **Roque, N.A.**, & Harrell, E.R. (2018). Registered replication report: Dijksterhuis & van Knippenberg (1998). *Perspectives on Psychological Science*, 1-38.
 43. **Roque, N. A.**, & Boot, W. R. (2018). A new tool for assessing mobile device proficiency in older adults: the mobile device proficiency questionnaire. *Journal of Applied Gerontology*, 37(2), 131-156, 0733464816642582. <https://doi.org/10.1177/0733464816642582>
 44. Dieciuc, M., **Roque, N. A.**, & Folstein, J. R. (2017). Changing similarity: Stable and flexible modulations of psychological dimensions. *Brain Research*, 1670, 208-219, 2017.06.026. <https://doi.org/10.1016/j.brainres.2017.06.026>
 45. Souders, D. J., Boot, W. R., Blocker, K., Vitale, T., **Roque, N. A.**, & Charness, N. (2017). Evidence for narrow transfer after short-term cognitive training in older adults. *Frontiers in Aging Neuroscience*, 9, 41, fnagi.2017.00041. <https://doi.org/10.3389/fnagi.2017.00041>
 46. **Roque, N.A.**, Wright, T. J., & Boot, W. R. (2016). Do different attention capture paradigms measure different types of capture?. *Attention, Perception, & Psychophysics*, 78, s13414-016-1117-4. <https://doi.org/10.3758/s13414-016-1117-4>
 47. **Roque, N.A.**, & Boot, W. R. (2015). Exploring the relationship between object realism and object-based attention effects. *Acta Psychologica*, 160, 178-183, 2015.07.014. <https://doi.org/10.1016/j.actpsy.2015.07.014> [**International Journal of Psychonomics; IF=1.380**]

Submitted

1. **Roque, N. A.** (submitted, The Gerontologist). Protecting ourselves from AI-enabled financial fraud: The case for the family password.
2. **Roque, N. A.**, Felt, J. (submitted, JMIR Research Protocols). Exploring risk factors for medication errors in daily life: A mixed methods study.

In Preparation

1. **Roque, N. A.**, & Shaleha, R. (in preparation). Screen time isn't all bad: A natural language analysis of how we measure it.
2. **Roque, N. A.**, & Shaleha, R. (in preparation). A taxonomy of screen-time interventions.
3. **Roque, N. A.**, Wang, D.H., Huncik, M., Santos, A. (in preparation). What one extra monitor can reveal: A case study on local air quality dynamics.
4. **Roque, N. A.**, Garcia de La Garza, A., Hosgood, H.D., Hall, C.B., Katz, M.J. (in preparation). Particulate matter 2.5 (PM2.5) air pollution exposure is associated with ambulatory cognitive assessment (ACA) performance: Results from the Einstein Aging Study.
5. **Roque, N. A.** (in preparation). Privacy and Safety in Online Research: Challenges and Opportunities. Target Journal: Behavior Research Methods. [**Premier Journal in Behavioral Research**]

Book Chapters

1. **Roque, N.A.**, & Germine, L.T. (2026). Digital Neuropsychology and the Assessment of Cognitive Change. Neuropsychology in the Age of Digital Health: Exploration, Application, and Practical Implementation.
2. Sliwinski, M.J., **Roque, N.A.**, Harrington, K. (2023). Cognition on the go: Opportunities and Challenges for Mobile Cognitive Health Research. Mobile Sensing in Psychology: Methods and Applications.
3. **Roque, N. A.**, & Boot, W. R. (2018). Action video games DO NOT promote visual attention. In Video game influences on aggression, cognition, and attention (pp. 105-118). Springer, Cham.
4. **Roque, N.A.**, Boot, W.R. (2017). Cognitive Aging. International Handbook of Positive Aging. London and New York: Routledge.

Technical Reports

1. Charness, N., Boot, W.R., Kaschak, M., Arpan, L., Cortese, J., Clayton, R., Stothart, C., **Roque, N.A.**, Paedae, B., & Barajas, K. (2017). Human Factors Guidelines to Develop Educational Tip Cards for Aging Road Users. Florida Dept. of Transportation Technical Report. https://rosap.ntl.bts.gov/view/dot/37111/dot_37111_DS1.pdf
2. Boot, W.R., Charness, N., Mitchum, A., **Roque, N.A.**, Stothart, C., & Barajas, K. (2015). Driving Simulator Studies of the Effectiveness of Countermeasures to Prevent Wrong-Way Crashes. Florida Dept. of Technical Transportation Report. https://rosap.ntl.bts.gov/view/dot/39272/dot_39272_DS1.pdf
3. Boot, W.R., Charness, N., **Roque, N.A.**, Barajas, K., Dirghalli, J., & Mitchum, A. (2015). The Flashing Right Turn Signal with Pedestrian Indication: Human Factors Studies to Understand the Potential of a New Signal to Increase Awareness of and Attention to Crossing Pedestrians. Florida Dept. of Transportation Technical Report. https://rosap.ntl.bts.gov/view/dot/29644/dot_29644_DS1.pdf

Conference Proceedings

1. Emihovich, B., **Roque, N.**, Criley, W., Criley, R., Criley, S., Alagoz, J., & Vukanovic-Criley, J. (2025). Increasing STEM Career Interest in Middle Schoolers Through a Health-Themed Serious Game. In Proceedings of the 18th European Conference on Game-Based Learning. Academic Conferences and Publishing Limited.
2. Flores-Cruz, G., *Hinkle, S.*, **Roque, N.**, & Mouloua, M. (2023). ChatGPT as the ultimate travel buddy or research assistant: a study on perceived attitudes and usability. Proceedings of the Human Factors and Ergonomics Society, 67.
3. Thai, P.*, Flores-Cruz, G., & **Roque, N.** (2023). Leveraging healthcare technology to improve patient-doctor communication. Proceedings of the Human Factors and Ergonomics Society, 67.
4. Boot, W. R., Souders, D., Charness, N., Blocker, K., **Roque, N. A.**, & Vitale, T. (2016). The gamification of cognitive training: Older adults' perceptions of and attitudes toward digital game-based interventions. In Human Aspects of IT for the Aged Population. Design for Aging (pp. 290-300). Springer International Publishing.
5. **Roque, N. A.**, Boot, W. R., Charness, N., Barajas, K., Dirghalli, J., & Mitchum, A. (2016). The Flashing Right Turn Signal with Pedestrian Indication: A human factors study to assess driver comprehension. In Human Aspects of IT for the Aged Population. Healthy and Active Aging (pp. 417-427). Springer International Publisher.

Research Funding

Current & Pending External Research Funding

Years	Role	Sponsor	Project	Total Costs
2026–2027	Co-Investigator	The Peanut Institute Foundation	Peanut butter, glycemic control, cognition, and cardiovascular health	\$290,000
2025–2030	Co-Investigator	NIH/NIA	THE NIH PRECISION BRAIN HEALTH NETWORK: Open and Inclusive Measurement of Neuropsychological Change for AD/ADRD Primary Prevention Trials (U24AG092760)	\$39,000,000*
2025–2030	Co-Investigator	NIH/NIA	Advancing Mobile Monitoring of Intraindividual Change in Alzheimer's Disease and Related Dementias: Building an Open and Inclusive Infrastructure (U19AG087932)	\$19,844,799*

Years	Role	Sponsor	Project	Total Costs
2023–2028	Subaward PI	NIH/NIA	Validation of the Remote CARE Toolbox for Diverse Populations (R01AG080635)	\$868,564
2022–2027	Co-PI	NIH/NIA	Einstein Aging Study: Air Pollution and Cognitive Function (P01AG003949)	\$911,768
2022–2027	Core Leader / Subaward PI	NIH/NIA	Einstein Aging Study: Technology & Data Management Core (P01AG003949)	\$1,696,559
2022–2024	Subaward PI	NIH/NIA	Perceived Ethnic Discrimination and Cognitive Function (R21AG078846)	\$90,232
2022–2023	Subaward PI	NIH/NIA	Mobile Monitoring of Cognitive Change (M2C2; U2CAG060408)	\$50,000
2021–2022	Subaward PI	Proactive Life, Inc.	Multi-modal Therapy to Improve Sleep and Cognition (R44AG056250)	\$14,685

Years	Role	Sponsor	Project	Total Costs
2020–2022	PI	NIH/NIA	Investigating Medication Errors in Daily Life (R03AG067009)	\$149,000

* Multi-project infrastructure award. Serves as Co-Core Leader for the Data Management & Analytics Core.

Funding Summary

- Principal Investigator on an NIH R03 award.
- Principal Investigator on four NIH subawards.
- Co-Principal Investigator on the Einstein Aging Study Program Project (P01) - Project 3, Air Quality and Cognition.
- Core Leader on the Einstein Aging Study Technology & Data Management Core.
- Co-Core Leader on recently funded NIH U24 and U19 infrastructure awards
- Co-Investigator on a foundation-funded clinical nutrition study.

Awards, Honors, & Distinctions

1. 2025 — Neurobiology and Changing Ecosystems Fellowship from the Research Corporation for Science Advancement
2. 2022 — UCF College of Sciences Excellence in Research and Creative Activities Celebration Award Recipient
3. 2017 — Mark & Gloria Charness Cognitive Science Award
4. 2017 — Florida State University, Leslie N. Wilson-Delores Auzenne Assistantship
5. 2016 — ASAP (Accessibility and Safety for an Aging Population) Doctoral Fellow Award
6. 2016 — Florida State University, Leslie N. Wilson-Delores Auzenne Assistantship
7. 2015 — Florida State University, Leslie N. Wilson-Delores Auzenne Assistantship
8. 2014 — NIMH Summer Institute in Cognitive Neuroscience Fellowship, UC Santa Barbara

Teaching

Pennsylvania State University

Term	Course / Role
Spring 2026	Adult Development and Aging
Fall 2025	No teaching, Course Buyout

Term	Course / Role
Spring 2025	Adult Development and Aging (Instructor of Record, co-teaching with A. Santos)
Fall 2024	No teaching, Course Release
Spring 2024	Adult Development and Aging; Individual Studies
Fall 2023	Individual Studies

University of Central Florida

Term	Course / Role
Summer 2023	Independent Study
Summer 2023	Honors Directed Readings II
Summer 2023	Directed Independent Research
Spring 2023	Advanced Human-Computer Interaction
Spring 2023	Directed Independent Research
Spring 2023	Honors Thesis I and II
Spring 2023	Independent Study
Fall 2022	Cognitive Psychology
Fall 2022	Honors Directed Readings I
Fall 2022	Directed Independent Research
Fall 2022	Honors Thesis I
Spring 2022	Advanced Human-Computer Interaction
Spring 2022	Honors Directed Readings
Spring 2022	Directed Independent Research
Fall 2021	Directed Independent Research
Spring 2021	Cognitive Psychology
Fall 2021	Cognitive Psychology

Florida State University

Term	Course / Role
Fall 2016	Brain and Behavior
Spring 2016	Research Methods
Summer 2014	Python Programming + Open-Source Experiment Building Workshop (Co-Lecturer)

Teaching Preparation and Training

Item	Details
Completed Teaching Practicum	FSU Psychology Department Teaching Practicum, Department of Psychology (Supervisors: Dr. Orenda Johnson, Dr. Justin Moss). January 2016–May 2016.
Professional Training	ABCs of Diversity (December 2021); Essentials of Online Teaching (Fall 2020); FERPA Training (May 2020).

Student and Postdoctoral Mentorship

Postdoctoral & Career Development (K Awards & Fellowships) Scholars

Mentee	Award	Years	Role	Status
Stephen Adamo	K99/R00	2023–Present	Consultant	Active
Jinshil Hyun	K99/R00	2023–Present	Consultant	Active

Ph.D. Dissertations Chaired

Student	Year	Department	University	Current Position
Rinanda Shaleha	2026	Human Development & Family Studies	Penn State	Assistant Professor, Indonesia University

Graduate Committee Membership (Non-Chair)

Student	Degree	Department	University	Role
Alexa Allan	Ph.D.	Human Development & Family Studies	Penn State	Committee Member
Jordan Sasser	Ph.D.	Psychology	University of Central Florida	Committee Member

Student	Degree	Department	University	Role
John Duany	Ph.D.	Psychology	University of Central Florida	Committee Member
Alexander Arca	Ph.D.	Psychology	University of Central Florida	Committee Member
Gabriela Flores-Cruz	Ph.D.	Psychology	University of Central Florida	Committee Member

Undergraduate Research & Honors Theses

- N=4 Successfully chaired Honors Theses at University of Central Florida
- N=1 successful honors thesis as committee member
- N=13 Undergraduate students supervised for research activities for 1 or more semesters.

Professional Service

University Service

Years	Activity
2024–2025	Member , Research Impacts Working Group, Social Science Research Institute, Penn State. Contribute metrics for measuring research impact.
2023–2024	Member , Search Committee, Environmental Health Science Initiative, College of Health and Human Development, Penn State. Participated in candidate screening, assessment, and interviews for two tenure-track positions with SSRI and ICDS affiliations.
2023–2025	Member , Professional Preparation, Development, and Evaluation (PPD&E) Committee, Department of Human Development and Family Studies, Penn State. Participate in data curation across HDFS programs, preparation of departmental identity surveys, committee discussions, and departmental planning.

Years	Activity
2025-2026	Member , Graduate Program Committee, Department of Human Development and Family Studies, Penn State. Active participant in planning of upcoming graduate program revisions and program enhancements.

Federal Funding Review

Year	Activity
2024	Early Career Reviewer , National Institutes of Health (NIH), Human Complex Mental Function (HCMF) Study Section. Reviewed grant applications and provided critiques and impact scores for funding consideration.
2023	Early Career Reviewer , National Institutes of Health (NIH), F02B Study Section. Reviewed grant applications and provided critiques and impact scores for funding consideration.

Professional Leadership

Years	Leadership Role
2024–2026	Program Chair , American Psychological Association (APA), Division 20 (Adult Development and Aging).
2014–2016	Graduate Student Representative , American Psychological Association (APA), Division 20 (Adult Development and Aging). Developed strategies to increase membership, produced newsletter content, and organized conference social events.

Professional Society Service

Years	Activity
2022, 2025, 2026	Early Career Award Reviewer , Society for Ambulatory Assessment (SAA). Reviewed applications for the Society for Ambulatory Assessment Early Career Award.
2018, 2019	Conference Abstract Reviewer , American Psychological Association (APA), Division 20 (Adult Development and Aging).
2020	Student Grant Reviewer , Department of Kinesiology, Penn State University. Participated in a mock NIH-style study section review.
2019	External Reviewer , Society for Research in Child Development (SRCD), Student and Early Career Council Dissertation Research Funding Awards.
2015	Poster Judge , Division 20 Poster Session, American Psychological Association Annual Meeting.

Editorial and Peer Review Activities

Ad Hoc Reviewer

- *Applied Ergonomics*
- *Frontiers in Digital Health*
- *International Journal of Psychophysiology*
- *Journal of Applied Gerontology*
- *Journal of Experimental Psychology: Applied*
- *Journal of Gerontology: Psychological Sciences*
- *Journal of Medical Internet Research (JMIR)*
- *JMIR Serious Games*
- *Journal of Visualized Experiments (JoVE)*

Professional Memberships

- Member, Gerontological Society of America
- Member, American Psychological Association, Division 20 (Adult Development and Aging)
- Member, American Psychological Association, Division 21 (Applied Experimental and Engineering Psychology)
- Member, American Psychological Association, Division 38 (Society for Health Psychology)

Media and Scientific Outreach

Selected National Media Coverage

- *The New York Times*
- *STAT*
- *ScienceDaily*
- *Ars Technica*
- *Boston Globe*
- CNN affiliates
- Phys.org
- *Inside Climate News*
- *EurekAlert!*
- *News-Medical*

Full Media Coverage List

Media for: Roque, N. A., Andrews, H., & Santos-Lozada, A. R. (2025). Identifying air quality monitoring deserts in the United States. *Proceedings of the National Academy of Sciences*, 122(17), e2425310122.

- Dzombak, Rebecca. 2025. “Almost Half of Americans Breathe Unhealthy Air, Report Finds.” *The New York Times*. <https://www.nytimes.com/2025/04/23/climate/american-lung-association-air-report.html>
- “Farming Through the Haze and Air Quality Concerns - Morning Ag Clips.” <https://www.morningagclips.com/farming-through-the-haze-and-air-quality-concerns/>
- Magazine, Undark. 2025. “Wildfires Are Challenging Air Quality Monitoring Infrastructure.” *Ars Technica*. <https://undark.org/2025/07/04/wildfires-aqi-infrastructure/>
- King, Jesse. 2025. “Bard College Expands Hudson Valley Air Monitoring Initiative, Details Findings in Kingston.” *WAMC*. <https://www.wamc.org/news/2025-06-30/bard-college-expands-hudson-valley-air-monitoring-initiative-details-findings-in-kingston>
- Kelly, Sharon. 2025. “As Trump Unwinds Federal Oversight, States Become Battlegrounds for Environmental Data.” *DeSmog*. <https://www.desmog.com/2025/06/11/as-trump-unwinds-federal-oversight-states-become-battlegrounds-for-environmental-data/>
- “Environmental Groups Sue over Louisiana’s Ban on Community Air Monitoring.” 2025. *Yahoo News*. <https://www.yahoo.com/news/environmental-groups-sue-over-louisiana-100057123.html>
- Newsource, C. N. N. 2025. “How Many Americans Are Breathing Unhealthy Air?” *KION Central Coast*. <https://kioncentralcoast.com/cnn-regional/2025/04/28/how-many-americans-are-breathing-unhealthy-air/>
- Newsource, C. N. N. 2025. “How Many Americans Are Breathing Unhealthy Air?” *KVIA*. <https://kvia.com/cnn-regional/2025/04/28/how-many-americans-are-breathing-unhealthy->

air/

- “Es seguro respirar el aire? 50 millones en EE. UU. en riesgo.” 2025. <https://noticias-delatierra.com/que-tan-seguro-es-respirar-el-aire-50-millones-de-personas-en-ee-uu-no-lo-saben/>
- Hawkins, J. 2025. “Almost Half of Americans Breathe Unclean Air, and It’s Only Getting Worse.” *BGR*. <https://www.bgr.com/science/almost-half-of-americans-breathe-unclean-air-and-its-only-getting-worse/>
- Douglas, Erin. 2025. “Air Quality Is Getting Worse in Coastal New England.” *Boston Globe* (Online). <https://www.proquest.com/docview/3193727979/citation/7CAC9A9C28804CD-CPQ/5>
- Lomte, Tarun Sai. 2025. “Study Pinpoints Vast ‘Air Quality Deserts’ across Nearly 60% of US Counties.” *News-Medical*. <https://www.news-medical.net/news/20250424/Study-pinpoints-vast-air-quality-deserts-across-nearly-6025-of-US-counties.aspx>
- Bennett, P. 2025. “Nearly Half of U.S. Population Breathes Unhealthy Air, American Lung Association Warns.” *EcoWatch*. <https://www.ecowatch.com/air-pollution-exposure-us-population-2025.html>
- Facher, Lev. 2025. “Over Half of U.S. Counties Have No System in Place for Monitoring Air Quality.” *STAT*. <https://www.statnews.com/2025/04/23/health-news-air-pollution-fda-food-dye-measles-texas-update-morning-rounds/>
- Gopal, Keerti. 2025. “Nearly Half of Americans Are Breathing Unhealthy Air as Pollution Exposure Numbers Reach Decade High.” *Inside Climate News*. <https://insideclimatenews.org/news/23042025/american-lung-association-unhealthy-air-report-pollution-exposure/>
- “Millions of Americans Live in Areas without Air Quality Monitoring.” *Earth.com*. <https://www.earth.com/news/millions-of-americans-live-in-areas-without-air-quality-monitoring/>
- Bioengineer. 2025. “Just How Safe Is the Air We Breathe? Find out What 50 Million Americans Are Missing.” *BIOENGINEER.ORG*. <https://bioengineer.org/just-how-safe-is-the-air-we-breathe-find-out-what-50-million-americans-are-missing/>
- “How Safe Is the Air to Breathe? 50 Million People in the US Do Not Know.” *EurekAlert!* <https://www.eurekalert.org/news-releases/1081150>
- “How Safe Is the Air to Breathe? 50 Million People in the US Do Not Know.” *ScienceDaily*. <https://www.sciencedaily.com/releases/2025/04/250421162612.htm>
- “Millions in the US Live without Local Air Quality Monitoring.” *MSN*. <https://www.msn.com/en-gb/news/world/millions-in-the-us-live-without-local-air-quality-monitoring/ar-AA1DlSKn>
- DNYUZ. 2025. “Almost Half of Americans Breathe Unhealthy Air, Report Finds.” <https://dnyuz.com/2025/04/23/almost-half-of-americans-breathe-unhealthy-air-report-finds/>
- “50 Million Live in U.S. Counties without Air Quality Monitoring.” *Laboratory Equipment*.

<https://www.laboratoryequipment.com/618975-50-Million-Live-in-U-S-Counties-without-Air-Quality-Monitoring/>

- News, Mirage. “50 Million Americans Unaware of Air Quality Safety.” *Mirage News*. <https://www.miragenews.com/50-million-americans-unaware-of-air-quality-1447394/>
- SCIENMAG. 2025. “Just How Safe Is the Air We Breathe? Find out What 50 Million Americans Are Missing.” *Science*. <https://scienmag.com/just-how-safe-is-the-air-we-breathe-find-out-what-50-million-americans-are-missing/>
- State, Penn. 2025. “Millions in the US Live without Local Air Quality Monitoring.” *News-Medical*. <https://www.news-medical.net/news/20250421/How-Safe-Is-The-Air-To-Breathe-50-Million-People-In-The-US-Do-Not-Know.aspx>
- Wagner, Aaron, and Pennsylvania State University. “How Safe Is the Air to Breathe? 50 Million People in the US Don’t Know.” <https://phys.org/news/2025-04-safe-air-million-people-dont.html>

Scientific and Professional Outreach

- May 2025. Stavanger, Norway. International Joint Programme - Neurodegenerative Disease Research (JPND) Working Group Lead; leading two writing groups in producing a guidelines paper and systematic review.
- 2015, 2018. Judge. Capital Regional Science and Engineering Fair, Tallahassee, FL. Judged student poster presentations and participated in award selection.

Public Commentary

- Penn State Outreach, featured Assistant Professor Nelson Roque, discussing healthy and unhealthy screen time during Screen Free Week. (n.d.). “Good and bad screen time for Screen Free Week | YouTube. <https://www.youtube.com/watch?v=Rr7892IxZQA>
- **Roque, N. A.**, provided commentary on the potential impact of fireworks on human health. (2022, December). “How Do Fireworks Affect the Air We Breathe? We Answer Key Questions about the Pollution They Generate.” <https://www.univision.com/noticias/fuegos-artificiales-contaminacion-aire-respiramos-preguntas-clave>
- **Roque, N. A.**, provided commentary on the state of detecting cognitive changes. (2019, December). “If Dementia Can’t Be Prevented, Can It at Least Be Detected Early?” <https://qz.com/1754544/detecting-dementia-early-could-the-key-to-a-cure/>
- **Roque, N. A.**, provided audio commentary on the state of assessing mobile device proficiency. (2018, December). “JAG: A New Tool for Assessing Mobile Device Proficiency in Older Adults: The Mobile Device Proficiency Questionnaire.” <http://sagenursing.sage-publications.libsnp.com/jag-a-new-tool-for-assessing-mobile-device-proficiency-in-older-adults-the-mobile-device-proficiency-questionnaire-0>
- **Roque, N. A.**, and Boot, W. R., provided audio commentary on the state of assessing mobile device proficiency. (2018, November). “JAG: A New Tool for Assessing Mobile

Device Proficiency in Older Adults: The Mobile Device Proficiency Questionnaire.”
<http://sagenursing.sage-publications.libsypnro.com/jag-a-new-tool-for-assessing-mobile-device-proficiency-in-older-adults-the-mobile-device-proficiency-questionnaire>

Community Outreach

2018–2019. Guest Lecturer. Pennsylvania CareerLink. Delivered public lectures on project management tools, digital work preparation, and practical strategies for remote work (“Project Management Tools for Efficiency and Success,” “Work from Anywhere: Preparing Your Digital Toolbox,” and “Work from Anywhere: Tools & Practical Advice”).

2017–2018. President. Tallahassee Apple Users Group Inc., 501(c)(3) non-profit organization. Led operations, developed community partnerships, and delivered technology-focused presentations.

2016–2017. Vice President. Tallahassee Apple Users Group Inc., 501(c)(3) non-profit organization. Delivered presentations on consumer technology and maintained organizational website.

Presentations

Invited Talks

1. **Roque, N.A.** (2025, May 20th). Detection of AI-Generated Information Across the Lifespan. In-person presentation for the IAFCI PA State Annual Conference in State College, PA.
2. **Roque, N.A.** (2025, March). Detection of AI-Generated Images Across the Lifespan. In-person presentation for the Mental Health Technology Center (MHTech) Seminar Series at the University of California San Diego.
3. **Roque, N.A.** (2024, May 10th). Breathe Well, Age Well. Virtual presentation as part of Stony Brook’s Cognitive Aging Colloquium.
4. **Roque, N.A.** (2024, January 22nd). Beyond Numbers: Unlocking Health Insights in Underserved Communities through mHealth Data Collection. In-person presentation and Q&A panel as part of UT Austin’s CAPS Panel on Ecological Momentary Assessments in Underrepresented Populations.
5. **Roque, N.A.** (2022, October 20th). Cognitive Assessment in the Cloud: Approaches & Lessons Learned. Virtual presentation as part of Amazon Web Services (AWS) Education Research Seminar Series.
6. **Roque, N.A.** (2022, October 6th). Cognitive Assessment in the Cloud: Approaches & Lessons Learned. In-person presentation at Union College. Schenectady, New York.
7. **Roque N.A.** (2020, August 20th). Leveraging Air Quality Sensors to Detect Smoking Behaviors. Virtual presentation at Pennsylvania State University x Monash University: Intersection of Health and the Environment.

Organized Symposia

1. Harrington, K., **Roque, N.**, & Mogle, J. (2020, November). Approaches for Modeling of Intensive Longitudinal Data to Understand Cognitive Aging. *Innovation in Aging*, 4(Suppl 1), 565. Gerontological Society of America, 2020 Annual Meeting. – Symposium Co-Chair.
2. Sliwinski, M.J. and **Roque, N.A.** (2020, April). Increasing the Precision of Measuring Cognitive Function in Older Adults using Mobile Technologies. *Cognitive Aging Conference*, 2020 Annual Meeting. – Symposium Co-Organizer; accepted; conference cancelled due to COVID-19.
3. Roque, NA., Hyun, J., Scott, S. (2019, November). Using Mobile Technology to Elucidate Psychosocial and Cognitive Function in Older Adults. *Innovation in Aging*, 3(Supplement_1), S809-S810. Gerontological Society of America, 2019 Annual Meeting. – Symposium Chair.

Conference Oral Presentations

1. Hyun, J., **Roque, N.**, Lovasi, G., Hosgood, D., Katz, M., Lipton, R., & Hall, C. (2025). Beyond Neighborhood Boundaries: GPS-Based Activity Space, Built Environment, and Cognitive Health. *Innovation in Aging*, 9(Supplement_2), igaf122.949. <https://doi.org/10.1093/geroni/igaf122.949>
2. De La Garza, A. G., Derby, C., Wang, C., Mogle, J., Lipton, R., **Roque, N.**, Katz, M., & Rabin, L. (2025). Perception to Performance: Linking Cognitive Lapses & Smartphone Cognitive Assessments in the Einstein Aging Study. *Innovation in Aging*, 9(Supplement_2), igaf122.1890. <https://doi.org/10.1093/geroni/igaf122.1890>
3. Shaleha, R., Almeida, D., & **Roque, N.** (2025). Daily Stress, Cortisol Dysregulation, and Screen Time in Midlife and Older Adults: Findings from the MIDUS Study. *Innovation in Aging*, 9(Supplement_2), igaf122.3065. <https://doi.org/10.1093/geroni/igaf122.3065>
4. Hoogendoorn, C., De La Garza, A. G., **Roque, N.**, Wang, C., Lipton, R., & Gonzalez, J. (2025). Wearable Continuous Glucose Monitoring and Cognitive Performance in Older Adults With Type 2 Diabetes. *Innovation in Aging*, 9(Supplement_2), igaf122.1889. <https://doi.org/10.1093/geroni/igaf122.1889>
5. Sergeyev, N., Bodek, H., Moffitt, C., Andrews, H. J., Cameron, J. D., Lavin, R., **Roque, N. A.**, Ezzati, A., Lipton, R. B., & Rabin, L. A. (2025, July 27). Impact of Sociodemographic and Linguistic Factors on TICS-m Screening: Findings from the Remote-CARE Study. *Alzheimer's Association International Conference*. <https://alz.confex.com/alz/2025/meetingapp.cgi/Paper/106205>
6. Go Cheng, K. J., & **Roque, N. A.** (2025, July 27). Hitting the Brakes: Driving Cessation, Cognitive Health, and the Mediating Role of Social Relationships Among Older Adults in the U.S. *Alzheimer's Association International Conference*. <https://alz.confex.com/alz/2025/meetingapp.cgi/Paper/099940>
7. Garcia De La Garza, A., Derby, C. A., Wang, C., **Roque, N. A.**, Katz, M. J., Lipton, R.

- B., & Rabin, L. A. (2025b, July 27). How Daily Experiences of Stress and Emotions Contextualize Subjective Cognitive Concerns: Findings from Daily Digital Diary Assessments in the Einstein Aging Study (EAS). Alzheimer’s Association International Conference. <https://alz.confex.com/alz/2025/meetingapp.cgi/Paper/107552>
8. Garcia De La Garza, A., Derby, C. A., Wang, C., **Roque, N.**, Katz, M. J., Lipton, R. B., & Rabin, L. A. (2025c, July 26). Elucidating the Association Between Pain and Self-Perceived Cognitive Functioning Using Daily Digital Diary Assessments: Results from the Einstein Aging Study (EAS). Alzheimer’s Association International Conference. <https://alz.confex.com/alz/2025/meetingapp.cgi/Paper/110543>
 9. Garcia De La Garza, A., Derby, C. A., Wang, C., **Roque, N. A.**, Katz, M. J., Lipton, R. B., & Rabin, L. A. (2025a, July 26). Longitudinal Increases in Subjective Cognitive Concerns are associated with Changes in Global Cognitive Performance: Findings from Daily Digital Diary Assessments in the Einstein Aging Study (EAS). Alzheimer’s Association International Conference. <https://alz.confex.com/alz/2025/meetingapp.cgi/Paper/110603>
 10. **Roque, N.**, Sliwinski, M., Katz, M., & Anderson-Hanley, C. (2023). Revolutionizing Cognitive Assessment: The Power of Digital Biomarkers and Ecologically Valid Methods. In-person oral presentation by lead-author at the Annual Scientific Meeting of the Gerontological Society of America (GSA).
 11. **Roque, N.A.** (2022, August). Exploring Model-based Profiles of Adherence to a Technology-Based Cognitive Intervention. Presented in-person at the American Psychological Association Annual Conference. [**Premier Psychology Conference**]
 12. **Roque, N.A.**, Adamo, S.*, Lago, M., Barufaldi, B. (2022, July). Modeling Quit-time Behavior in a Satisfaction of Search (SOS) Experimental Framework: The Role of Context and Experience. Presented virtually at the Medical Image Perception Society Annual Conference.
 13. **Roque, N.A.** (2022, April). Implementation of Web-based Experiments: Considerations for a Live Study. Presented virtually at the Cognitive Aging Conference.
 14. Harrell, E., **Roque, N.**, Boot, W., & Charness, N. (2021). Exploring Factors that Shape Adherence to Technology-Based Cognitive Interventions. *Innovation in Aging*, 5(Suppl 1), 843-843. [**Premier Gerontology Conference**]
 15. **Roque, N. A.**, Hall, C. B., Katz, M., & Sliwinski, M. J. (2021, November). Air Pollution Exposure is Associated with Cognitive Performance: Results from the Einstein Aging Study. Virtual presentation by lead-author at the 2021 Annual Scientific Meeting of the Gerontological Society of America (GSA) conference. *Innovation in Aging*, 5(Suppl 1), 138-138. [**Premier Gerontology Conference**]
 16. **Roque, N. A.**, Hall, C. B., Katz, M., & Sliwinski, M. J. (2021, August). Air Pollution Exposure is Associated with Cognitive Performance: Results from the Einstein Aging Study. Virtual presentation by lead-author at the 2021 International Society for Environmental Epidemiology (ISEE) conference. In *ISEE Conference Abstracts* (Vol. 2021, No. 1).

[Premier International Environmental Epidemiology Conference]

17. Boot, W., **Roque, N.**, Harrell, E., & Charness, N. (2020). Older Adults' Adherence to Technology-Based Intervention: The Role of Messaging and Individual Differences. Virtual presentation by lead-author. *Innovation in Aging*, 4(Suppl 1), 555.
18. Charness, N., Souders, D., Best, R., **Roque, N.**, Yoon, J., & Stothart, C. (2020). Acceptance of Transportation Technologies by Aging Adults. Virtual presentation by lead-author. *Innovation in Aging*, 4(Supplement_1), 555-555.
19. Harrington, K., **Roque, N.**, Zhaoyang, R., Mogle, J., & Sliwinski, M. (2020). Loneliness and Profiles of Objective and Subjective Memory During Midlife. Virtual presentation by lead-author. *Innovation in Aging*, 4(Suppl 1), 565.
20. Cerino, E. S., Hyun, J., Hakun, J. G., **Roque, N. A.**, Lipton, R. B., & Sliwinski, M. J. (2020). Variability in working memory performance on mobile devices is sensitive to mild cognitive impairment: Results from the Einstein Aging Study: Neuropsychology/early detection of cognitive decline with neuropsychological tests. Virtual presentation by lead-author. *Alzheimer's & Dementia*, 16, e039960.
21. Sliwinski, M. J., Hakun, J. G., **Roque, N.**, Katz, M. J., Zhaoyang, R., & Lipton, R. B. (2019). Intensive ambulatory cognitive assessments in older adults: sensitivity to cognitive risk status in the Einstein Aging Study (EAS). Virtual presentation by lead-author. *Alzheimer's & Dementia*, 15(7), P355-P356.
22. Oravec, Z., **Roque, N.**, & Sliwinski, M. (2020, November). Bayesian Modeling of Cognitive Impairment in the Presence of Retest Effects. *Innovation in Aging*, 4(Suppl 1), 566. Virtual presentation presented by lead-author at the Annual Scientific Meeting of the Gerontological Society of America (GSA). Virtual
23. **Roque, N.**, Sliwinski, M., & Ram, N. (2020, November). Questionnaire-Based Everyday Reaction Time: Reliable, Valid, and Unobtrusive Measure of Cognition. *Innovation in Aging*, 4(Suppl 1), 599. Virtual presentation by lead-author at Annual Scientific Meeting of the Gerontological Society of America (GSA).
24. **Roque, N.**, Ram, N., & Ong, A. (2019, January). Predicting health outcomes from time series features of affect dynamics: a machine learning approach. Presented by lead-author at the Bi-Annual World Congress Meeting of the International College of Psychosomatic Medicine (ICPM). Florence, Italy. In *Psychotherapy and Psychosomatics* (vol. 88, pp. 110-110). Allschwilerstrasse 10, CH-4009 Basel, Switzerland: Karger. <https://www.karger.com/Article/Pdf/502467>
25. Ong, A., **Roque, N.**, & Ram, N. (2019). Fragile positive affect and systemic inflammation. Presented by lead-author at the Bi-Annual World Congress Meeting of the International College of Psychosomatic Medicine (ICPM). Florence, Italy. In *Psychotherapy and Psychosomatics* (Vol. 88, pp. 100-100). Allschwilerstrasse 10, CH-4009 Basel, Switzerland: Karger. <https://www.karger.com/Article/Pdf/502467>
26. Mason, J., Jaime, S., **Roque, N.A.**, Maharaj, A., Tenenbaum, G., & Figueroa, A. (2019,

- May). The effects of an acute bout of aerobic exercise on cognitive function in older adults. Presented by lead-author at the Annual Meeting of the American College of Sports Medicine (ACSM). Orlando, FL.
27. **Roque, N.A.**, Boot, W.R. (2018, May). The Development and Validation of the Mobile Device Proficiency Questionnaire in U.S. Samples. Presented at the 2018 Annual Meeting of the International Society for Gerontechnology. St. Petersburg, FL.
 28. Souders, D.J., Charness, N., **Roque, N.A.**, Pham, H. (2018, April). Do Advanced Driver Assistance Systems (ADAS) improve older driver performance? Presented by lead-author at SUNY Stony Brook, Smart and Connected Community Conference. New York.
 29. **Roque, N.A.**, Boot, W.R., Charness, N., Barajas, K., Dirghalli, J., Mitchum, A. (2016, July). Exploring the Comprehension of a New Traffic Signal to Promote the Safety and Awareness of Pedestrians. Presented at Human Computer Interaction International (HCII). Toronto.

Conference Posters

1. Prieto, S., Split, M., Vito, A. N. D., Strenger, J., **Roque, N. A.**, Sliwinski, M. J., Thompson, L. I., & Kunicki, Z. J. (2025). The Impact of Stress on Cognitive Performance in Remote Smartphone-Based Assessments: Exploring Within-Person Variability. *Alzheimer's & Dementia*, 21(S3), e102501. https://doi.org/10.1002/alz70857_102501
2. Sergeyev, N., Bodek, H., Moffitt, C., Andrews, H. J., Cameron, J. D., Lavin, R., **Roque, N. A.**, Ezzati, A., Lipton, R. B., & Rabin, L. A. (2025). Impact of Sociodemographic and Linguistic Factors on TICS-m Screening: Findings from the Remote-CARE Study. *Alzheimer's & Dementia*, 21(S3), e106205. https://doi.org/10.1002/alz70857_106205
3. **Roque, N. A.**, Katz, M. J., Garza, A. G. D. L., Hosgood, D., & Hall, C. B. (2025). Exposures to PM2.5 in Older Adults: Differences by Cognitive Status and Age in the Einstein Aging Study. *Alzheimer's & Dementia*, 21(S2), e106788. https://doi.org/10.1002/alz70856_106788
4. Cheng, K. J. G., & **Roque, N. A.** (2025). Hitting the Brakes: Driving Cessation, Cognitive Health, and the Mediating Role of Social Relationships Among Older Adults in the US. *Alzheimer's & Dementia*, 21(S4), e099940. https://doi.org/10.1002/alz70858_099940
5. Anderson-Hanley, C., **Roque, N. A.**, Rogers, J., Needham, V., Panos, S., & Warner, G. (2025). Digital biomarker for assessing executive function in a remote iPACES clinical trial for mild cognitive impairment: The comparability of the electronic Stroop vs. the paper Stroop. *Alzheimer's & Dementia*, 21(S3), e107788. https://doi.org/10.1002/alz70857_107788
6. Andrews, H. J., Ezzati, A., Rabin, L. A., Sergeyev, N., Bodek, H., Cameron, J. D., Moffitt, C., Lavin, R., Lipton, R. B., & **Roque, N. A.** (2025). Open-Source Development of The Remote Cognitive Aging and Alzheimer's Disease Research R-CARE Toolbox for Advancing Remote Neuropsychological Assessment for Cohorts and Clinical Trials.

- Alzheimer's & Dementia, 21(S2), e106978. https://doi.org/10.1002/alz70856_106978
7. Garza, A. G. D. L., Derby, C. A., Wang, C., **Roque, N. A.**, Katz, M. J., Lipton, R. B., & Rabin, L. A. (2025). Elucidating the Association Between Pain and Self-Perceived Cognitive Functioning Using Daily Digital Diary Assessments: Results from the Einstein Aging Study (EAS). *Alzheimer's & Dementia*, 21(Suppl 9), e110543. https://doi.org/10.1002/alz70863_110543
 8. Velazquez, E., & **Roque, N.** (2025). Impact of Global and Local Clutter on Visual Search Efficiency and Attentional Guidance. *Journal of Vision*, 25(9), 2209–2209. <https://doi.org/10.1167/jov.25.9.2209>
 9. Nester, C.O., De Vito, A.N., Kunicki, Z.J., Strenger, J., Harrington, K.D., **Roque, N.A.**, Sliwinski, M.J., & Thompson, L.I. Subjective Cognitive Concerns and High Frequency Mobile App Based Cognitive Assessment in Older Adults: Relationship to Adherence, Satisfaction, and Cognitive Performance. Poster accepted for the Alzheimer's Association International Conference 2024. [**Premier International Alzheimer's Conference**]
 10. **Roque, N.A.**, Sliwinski, M., Derby, C., Lipton, R.B., Buxton, O., & Katz, M. (2023). Feasibility of A Multi-Pronged, Multi-Sensor Cognitive Aging Protocol in a Diversified Cohort: Results from the Einstein Aging Study. Poster accepted for the Alzheimer's Association International Conference 2024. [**Premier International Alzheimer's Conference**]
 11. Flores-Cruz, G.* & **Roque, N.** (2024, May). Eye Tracking Impacts of an Irrelevant Self-View Singleton. Poster accepted at the 2024 Vision Science Society Conference. To be presented by lead author.
 12. Velasquez, E.* & **Roque, N.** (2024, May). Assessing The Effect of Stimuli Complexity in Web-Based Visual Foraging. Poster accepted at the 2024 Vision Science Society Conference. To be presented by lead author.
 13. De Vito, A., Kunicki, Z., Britton, K., Villa, P., Strenger, J., Harrington, K., ... & Thompson, L. I. (2023). 5 Intraindividual Variability in Processing Speed on Digital Cognitive Assessments Differs by Amyloidosis Status in Cognitively Normal Older Adults. *Journal of the International Neuropsychological Society*, 29(s1), 217-218. [**Premier International Neuropsychology Conference**]
 14. Strenger, J. R., Riera, N., Harrington, K., **Roque, N.**, Salloway, S., Sliwinski, M., & Thompson, L. I. (2023). 28 Variability in Remote, Self-Administered Assessment Performance Associated with Self-Reported Memory Perceptions Among Older Adults. *Journal of the International Neuropsychological Society*, 29(s1), 340-341. [**Premier International Neuropsychology Conference**]
 15. DeBonis, D. G., Dunnam, M., Panos, S., Warner, G., **Roque, N.**, & Anderson-Hanley, C. (2023). 78 Examining the Association Between a Patient's Diagnosis and Occurrence of The First Error on Trails B. *Journal of the International Neuropsychological Society*, 29(s1), 280-280. [**Premier International Neuropsychology Conference**]

16. De Vito, A. N., Kunicki, Z.J., Emrani, S., Strenger, J., Harrington, K.D., **Roque, N.A.**, ... & Thompson, L. I. (2023). Fatigue Interacts with Amyloid Status To Predict Daily Working Memory Performance in Preclinical Alzheimer's Disease. *Alzheimer's & Dementia*, 19, e080094. [**Premier International Alzheimer's Conference**]
17. Strenger, J., Emrani, S., Kunicki, Z.J., Harrington, K.D., **Roque, N. A.**, Sliwinski, M.J., ... & Thompson, L. I. (2023). Time of day cognitive performance differences in remote, self-administered assessments and its association with cerebral amyloidosis. *Alzheimer's & Dementia*, 19, e080096. [**Premier International Alzheimer's Conference**]
18. Katz, M.J., De La Garza, A., **Roque, N.A.**, Derby, C.A., Sliwinski, M.J., Lipton, R.B., Wang, C. (2023). Feasibility, reliability and predictive validity of digital cognitive assessments in a diverse cohort of community-residing older adults. Presented in person by lead author at the 2023 Alzheimer's Association International Conference. [**Premier International Alzheimer's Conference**]
19. Harrington, E., Graham-Engeland, J.E., Lipton, R.B., **Roque, N.A.**, Sliwinski, M.J., Engeland, C.G. (2023). Inflammation and working memory performance in everyday life: Gender differences among older adults with and without cognitive impairment. Presented in person by lead author at the 2023 Alzheimer's Association International Conference. [**Premier International Alzheimer's Conference**]
20. Thai, P., *Flores-Cruz, G.*, & **Roque, N.** (2023, October). Leveraging healthcare technology to improve patient-doctor communication. Poster presented by lead author at the 67th annual conference of the Human Factors and Ergonomics Society. Conference proceeding posted in the Proceedings of the Human Factors and Ergonomics Society, 67.
21. Flores-Cruz, G., *Hinkle, S.*, **Roque, N.**, & Mouloua, M. (2023). ChatGPT as the ultimate travel buddy or research assistant: a study on perceived attitudes and usability. Conference proceeding posted in the Proceedings of the Human Factors and Ergonomics Society, 67.
22. Flores-Cruz, G.*, **Roque, N.** (2023, June). Perception of Environmental Risk: Comparing Mental Models of Indoor and Outdoor Air Pollution. Poster presented at the 35th annual conference of the International Society for Environmental Epidemiology.
23. **Roque, N.**, & Adamo, S.* (2023). Modeling Observer Search Termination in a Subsequent Search Misses (SSM) Experimental Framework: The Role of Experience. Poster presented at the 23rd Annual conference of the Vision Sciences Society. *Journal of Vision*, 23(9), 5936-5936.
24. **Roque, N.A.**, Ferguson, I.*, Saulnier, E.T., Anderson-Hanley, C. Trail Making for electronic touchscreen self-administration (eTrails) as used in the remote iPACES neuro-exergaming trial for Mild Cognitive Impairment (MCI): An initial validation report. Submitted to Alzheimer's Association International Conference 2022. Presented Virtually. [**Premier International Alzheimer's Conference**]
25. **Roque, N.A.**, Anderson-Hanley, C. Saulnier, E.T. (2021, October). Translating the Paper Stroop Task for Repeated Administration as an Electronic Stroop (eStroop) Task

- on Mobile Devices for a Remote Exercise Intervention for Mild Cognitive Impairment (MCI). (virtual) 11th edition of the International Conference on Frailty and Sarcopenia Research (ICFSR 2021).
26. Dalterio, C., *Ferguson, I.*, Nath, K., **Roque, N.A.**, Arciero, P., Anderson-Hanley, C. (2021, October). Biomarker and cognitive effects of single bouts of moderate-intensity neuro-exergaming, high-intensity interval training (HIIT), and fasted exercise for mild cognitive impairment: A case study of serial interventions. Presented by lead author at the (virtual) 11th edition of the International Conference on Frailty and Sarcopenia Research (ICFSR 2021).
 27. Harrell, E., **Roque, N.**, Boot, W., & Charness, N. (2021). Exploring Factors that Shape Adherence to Technology-Based Cognitive Interventions. *Innovation in Aging*, 5(Suppl 1), 843-843. Presented by the lead author at the (virtual) Annual Scientific Meeting of the Gerontological Society of America (GSA) [**Premier Gerontology Conference**]
 28. **Roque, N.**, & Harrell, E. (2020, November). Give People What They Want, Feedback: Older Adult Perceptions on Cognitive Training Features to Increase Engagement. *Innovation in Aging*, 4(Suppl 1), 411. Presented at the (virtual) Annual Scientific Meeting of the Gerontological Society of America (GSA). [**Premier Gerontology Conference**]
 29. Harrell, E., & **Roque, N.** (2020, November). I'd Rather Be...: Activities Older Adults Reported Leading to Decreased Engagement With Cognitive Training. *Innovation in Aging*, 4(Suppl 1), 325. Presented by the lead author at the (virtual) Annual Scientific Meeting of the Gerontological Society of America (GSA). [**Premier Gerontology Conference**]
 30. **Roque, N.**, & Harrell, E. (2020, October). Give the People What they Want, Feedback: Older adult perceptions of cognitive training features. *Innovation in Aging*, 4(Suppl 1), 411. Presented at the (virtual) Annual Scientific Meeting of the International Society for Gerontechnology (ISG). *Gerontechnology*, 19(0), 1-1. <https://doi.org/10.4017/gt.2020.19.s.70037> [**Premier Gerontology Conference**]
 31. **Roque N.A.**, Boot, W.R., (2018, November). A New Scale for Assessing Older Adults' Wireless Network Proficiency: The Wireless Network Proficiency Questionnaire. Presented at the Annual Scientific Meeting of the Gerontological Society of America (GSA). Boston, MA. [**Premier Gerontology Conference**]
 32. Lebeau, J.C., Mason, J., **Roque, N.A.**, Razon, S., & Tenenbaum, G. (2018, June). The Effects of Acute Exercise on Driving Performance and Executive Functions in Healthy Older Adults. Presented at North American Society for Psychology of Sport and Physical Activity's 51st Annual Meeting in Denver, Colorado.
 33. Mason, J., **Roque N.A.**, Maharaj A., Salvador, J., Figueroa, A. (2018, June). Exploring the effects of life-long aerobic exercise on cognitive function, driving performance, and cardiovascular health. Poster presented at American College of Sports Medicine's 65th Annual Meeting in Minneapolis, Minnesota.
 34. **Roque N.A.**, Boot, W.R., (2018, May). Pupillometry as a window into the content

- and strength of attention sets. Presented at the 18th Annual Meeting of the Vision Sciences Society. St. Petersburg Beach, FL. <https://jov.arvojournals.org/article.aspx?articleid=2700102>
35. Andringa, R., **Roque N.A.**, Boot, W.R., (2018, May). Exploring the Effect of Task Complexity, Presentation, and Input Device on the Near-Hands Advantage. Presented at the 18th Annual Meeting of the Vision Sciences Society. St. Petersburg Beach, FL. <https://jov.arvojournals.org/article.aspx?articleid=2699638>
 36. **Roque N.A.**, Harrell, E., Wright, T.J., Boot, W.R., (2018, May). Exploring Visual Search as a Paradigm for Predicting Medication Errors. Presented at Cognitive Aging Conference 2018. Atlanta, GA.
 37. Ospina J.*, Yazbec, A., **Roque, N.A.**, Borovsky, A. (2017, April). Pupillometry measures highlight the timing of cognitive effort during linguistic adaptation. Presented at CUNY 2017 Conference on Human Sentence Processing.
 38. **Roque N.A.**, Wright T.J., Boot W.R. (2016, May) Do different attention capture paradigms measure different types of capture? Presented at the 16th Annual Meeting of the Vision Sciences Society. St. Petersburg Beach, FL. <https://jov.arvojournals.org/article.aspx?articleid=2550984>
 39. **Roque N.A.**, Boot W.R. (2016, April) Mobile Device Proficiency Questionnaire (MDPQ): Assessing mobile smart device proficiency among seniors. Presented at Cognitive Aging Conference 2016. Atlanta, GA.
 40. **Roque N.A.**, Boot W.R. (2015, November). Do Different Attention Capture Paradigms Measure Different Types of Capture? Presented at the 23rd Object Perception, Attention and Memory (OPAM) Workshop. Chicago, IL.
 41. **Roque N.A.**, Wright T.J., Boot W.R. (2015, May). Exploring Visual Search as a Paradigm to Study Medication Errors. Presented at the 15th Annual Meeting of the Vision Sciences Society. St. Petersburg Beach, FL. <https://jov.arvojournals.org/article.aspx?articleid=2433979>
 42. **Roque N.A.**, Boot, W.R. (2014, May). Exploring the Relationship between Object-Based Attention Effects and Object Realism. Poster presented at the 14th Annual Meeting of the Vision Sciences Society. St. Petersburg Beach, FL. <https://jov.arvojournals.org/article.aspx?articleid=2144943>
 43. Driscoll K.A., Johnson S.B., Gill E., Pizzi A., **Roque N.A.**, Sorrentino E., ... (2012, August) "Grading" Type 1 Diabetes Knowledge: How Well Do Children and Parents Perform? Presented at the 120th Annual Convention of the American Psychological Association. Orlando, FL.

University / Department / Center: Oral Presentations

1. **Roque N.A.** (2023, October). Breathe Well, Age Well: A Comprehensive Look at Air Quality for Older Adults. Presented at the Penn State Center for Healthy Aging Colloquium.
2. **Roque N.A.** (2023, March). Reproducible Data Science. Workshop presentation for UCF Research IT. Workshop event listing: <https://events.ucf.edu/event/2891771/reproducible-data-science/>
3. **Roque N.A.** (2023, March). Qualtrics for Research Data Collection. Workshop presentation for UCF Research IT. Workshop event listing: <https://events.ucf.edu/event/2819983/qualtrics-for-research-data-collection/>
4. **Roque N.A.** (2021, September). Introduction to Data Wrangling & Visualization in R: Exploring the Google Mobility Dataset. Workshop presentation for UCF HFC Doctoral Program Brown bag series. Workshop materials open-sourced: https://github.com/nelsonroque/contextlab_intro_datascience
5. **Roque N.A.** (2019, November). Using Model-Based Cluster Analyses of Cognitive Performance to Unpack Differences in Psychosocial Risk Factors. Presentation for Pennsylvania State University's Center for Health Aging, Aging SIG.
6. **Roque N.A.** (2018, March). tsfeaturex: Automating the Extraction of Time Series Features. Presentation for Pennsylvania State University's QuantDev Brown Bag Lecture Series.
7. **Roque N.A.** (2018, April). Reliably measuring attention capture: Challenges and new approaches. Presentation for FSU Psychology's Graduate Research Day.
8. **Roque N.A.** (2017, April). Exploring visual search as a paradigm for predicting medication errors. Presentation for FSU Psychology's Graduate Research Day.
9. **Roque N.A.** (2016, April). Do different attention capture paradigms measure different types of capture? Presentation for FSU Psychology's Graduate Research Day.
10. **Roque N.A.** (2016, March). Exploring visual search as a paradigm for studying medication errors. Research sharing luncheon held by the FSU Fellows Society.